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24BCE0403

CAT-1 (CD1+TD1)
WINTER - SEM

Q.1 (a) (i) error
→ Comma not used
Correct : MUL AB

(ii) Error
→ push accept only
direct address
Correct : Push Acc

(iii) Error
→ Destination must be Accumulator
or direct Address

Correct : ANL A, R0

(iv) Error
→ R2-R7 Cannot be used as pointers, only
R0 and R1 supports indirect Addressing.

Correct : MOV A, @R0

(v) Error
→ Register to Register moves not
allowed

Correct : MOV A, R1
MOV R2, A

- 1(b) (i) Direct Addressing Mode
 (ii) Immediate Addressing Mode
 (iii) Register Addressing
 (iv) Indexed Addressing
 (v) Register Indirect Addressing

Qwe-2 ORG: 000H
 LJMP MAIN
 ORG 0030H

MAIN:

MOV DPTR, #250H;
 MOV R0, #40H;
 MOV R2, #06H;

LOAD - DATA:

CLR A;

MOVC A, @A + DPTR

MOV @R0, A;

INC DPTR;

INC R0;

DJNZ R2, LOAD-DATA

MOV R0, #40H;

MOV R2, #05H;

MOV A, @R0;

MOV R4, @R0;

ADD LOOP

INC R0;

ADD A, @R0;

DA A;

INC NEXT;

INC R4;

NEXT :

DJNZ R1, ADD-LOOP

MOV R3, A

H : SJMP H ;

ORG 250H

DATA-1 : DB 54H, 76H, 65H, 98H, 88H, 99H

END

Que-4

operations	Stack content	SP
push 0	0BH : 30H	08H
push 1	09H : 23H	09H
push 2	0AH : 00H	0AH
push 1	08H : 24H	0BH
pop 3	24H Removed	0AH
pop 7	00H Removed	09H
push 3	0AH : 24H	0AH
pop 6	24H Removed	09H

Reg/location

value in Hex

R0

30H

R1

24H

R3

24H

R7

00H

R6

24H

Qwe-5 (a) SETB PSW.4 $\Rightarrow$ Sets bit 4 of PSW
 MOV R2, #78H (RS1=1)
 MOV A, R2 $\Rightarrow$ Sets bit 3 of PSW
 SETB PSW.3 (RS0=1)
 MOV R4, #0A4H
 ADD A, R4
 END

Since RS1=1 & RS0=1 Hence reg bank 3 is selected.

$\rightarrow$ the data 78H is stored in R2 of bank 3. Ram address for R2 in bank 3 is 1AH

$\rightarrow$ the data A4H is stored in R4 of bank 3. The RAM address for R4 is 1CH.

(b)(i) 78H = 01111000
 A4H = 10100100
 SUM = 00011100 = 1CH with carry out
 ACC(A) = 1CH
 CY (carry flag) = 1
 AC (Auxiliary carry) = 0
 OR (overflow flag) = 1
 P (parity flag) = 1

(ii) operation	Accumulator
ANL A, #00H	00H
MOV A, #3AH	3AH
SWAP A	A3H
ANL A, #23H	23H
DIV AB	08H

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24BCE0403

CAI-1

SLOT :- D2 + TD2

WINTER - SEM

Qwe-1 (A) (i) Error

→ Can't move single bit into 8 bit accumulator

Correct : MOV A, P0

(ii) Error

→ decrement data pointer doesn't exist

Correct : INC DPTR

(iii) Error

→ Can't increment an immediate value

Correct : INC 40H

(iv) SETB A is not correct

→ SETB is bit Addressable

Correct : MOV A, #040H

(v) Error

→ CLR operation is bit addressable

Correct : CLR P2.0

$$(b) \quad T = \frac{12}{F} = \frac{12}{15 \times 10^6} = 0.8 \text{ MS}$$

Execution	Machine Cycle	Execution
MOV R3, #80H	1	1
HERE: NOP	1	80 (A)
NOP	1	80
NOP	1	80
NOP	1	80
DJNZ R3, HERE	2	80
RET	2	1

$$\Rightarrow \text{Loop body} = (1+1+1+1+2) \times 80$$

$$= 6 \times 80 = 480 \text{ Cycles}$$

$$\text{Total} = 1 + 480 + 2 = 483 \text{ Cycles}$$

$$\text{Total Execution Time} = 483 \times 0.8$$

$$= 386.4 \text{ MS}$$

Que-2

ORG 0000H ; A

START: MOV A, P1 ;

JB ACC.7, SQUARE-25 ;

WAVE-60: SETB P2.1 ;

ACALL DELAY-3 ;

CLR P2.1 ;

ACALL DELAY-2 ;

SJMP START


```
SETB P2.1 ;  
ACALL DELAY-1 ;  
CLR P2.1 ;  
ACALL DELAY-3 ;  
STMP START
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DELAY-3 : MOV R0, # 30H ;  
D3 : DJNZ R0, D3  
RET ;
```

```
DELAY-2 : MOV R0, # 20  
D2 : DJNZ R0, D2  
RET ;
```

```
DELAY-1 : MOV R0, # 10 ;  
D1 : DJNZ R0, D1 ;  
RET ;
```

```
END
```

Que-3

```
ORG 0000H  
MOV DPTR, # 300H  
MOV R1, # 10 ;  
MOV R2, # 00H ;  
MOV R3, # 00H ;  
BACK : CLR A ;  
MOVC A, @ A + DPTR ;  
ADD A, R2 ;  
MOV R2, A ;  
JNC NEXT ;  
INC R3 ;  
NEXT : INC DPTR ;  
DJNZ R1, BACK ;
```



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HERE : STMP HERE ;
      ORG 300H ;
      DB 92H, 34H, 84H, 29H, 22H, 11H,
        33H, 44H, 55H, 66H ;
      END

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Que-4	operation	Data in Rom	SP
	MOV 05H, #32H;	05H : 3EH	07H
	MOV 0AH, #DFH;	0AH : DFH	07H
	MOV 2FH, #45H;	2FH : 45H	07H
	MOV SP, #69H;	—	69H
	PUSH 05H;	6AH : 3EH	6AH
	PUSH 0AH;	6BH : DFH	BBH
	PUSH 2FH;	6CH : 45H	6CH
	POP 4AH;	4AH : 45H	6BH
	PUSH 4BH;	68H : DFH	6AH
	POP 5AH;	5AH : DFH	6BH

Que-5	(1) Instruction	Accumulator	C	RAM[20H]
	MOV A, #83H	83H (1000 0011)	0	—
	RCC A	06H (0000 0110)	1	—
	RLC A	0DH (0000 1101)	0	—
	MOV 20H, #87H	0DH	0	87H
	ANL C, D2H	0DH	0	87H

→ final Result :-

A : 0DH
 RAM[20H] : 87H
 Carry flag : 0

(ii)	Instruction	Reg A	Reg RO	RAM[40H]	Carry
MOV	A, #80H	80H	—	—	0
MOV	RO, #40H	80H	40H	—	0
MOV	@RO, A	80H	40H	80H	0
	INC A	81H	40H	80H	0
	RRC A	40H	40H	80H	1
	ADD A, 40H	C0H	40H	80H	0

→ final Result :-

A : C0H
 RO : 40H
 RAM[40H] : 80H
 Cy : 0